# PeakMeister 2.0

## server.R — Task Boundaries and GUI Locations

Revised task-boundary inventory: all 261 original rows and line ranges retained, with a GUI location / trigger column added.

Code baseline: 709b1bb0997c28519ed8eb144c43ffe62e553f7b. server.R has 6,096 lines; ui.R has 546 lines. Documentation updated 19 October 2026. No application source was changed for this documentation update.

*ui.R: sidebar, tabs and controls* | *server.R: task implementation*

Reading the GUI column: “>” follows sidebar page → panel/tab → control. “Shared controls” is the control column inside a page, separate from the main sidebar. Dialogs are opened by buttons. “Background” means there is no separate visible panel for that step. Shared helpers list multiple consumers. Legacy/uncalled helpers are explicitly marked.

The GUI guide, PeakMeister-GUI-guide.pdf, describes each panel’s inputs and outputs in detail. This document is a code-location index based on source inspection; it does not claim end-to-end validation of every GUI feature. Line ranges are inclusive and anchored to the baseline commit, not later documentation commits. Overlapping ranges represent subtasks.

## 1. Application setup and About tab

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Configure upload-size limit and timeout | 15 | 18 | All pages > application upload/timeout configuration (background) |
| Initialize stored Plotly objects | 23 | 24 | Engine > Initialize Run / Visualization > saved Plotly collection (background) |
| Initialize integration boundary positions | 26 | 27 | Visualization > Plotting > red/blue integration boundaries |
| Display disclaimer | 31 | 33 | About > Disclaimer |
| Render and display README | 34 | 36 | About > README |
| Display update history | 37 | 39 | About > Updates |
| Display license | 40 | 42 | About > License |

*The enclosing server() function runs from lines 21–6095.*

## 2. User-supplied parameters

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Initialize metabolite mass-list state | 49 | 62 | User Supplied Parameters > Mass List |
| Initialize reference-mass state | 65 | 77 | User Supplied Parameters > Reference Mass List |
| Initialize processing-parameter state | 80 | 91 | User Supplied Parameters > Parameters |
| Initialize user-supplied migration-index state | 95 | 101 | User Supplied Parameters > User Supplied Migration Indexes |
| Register initial parameter-table displays | 104 | 107 | User Supplied Parameters > all four data tables |
| Read the three worksheets from uploaded parameter Excel file | 111 | 128 | User Supplied Parameters > Mass List / Reference Mass List / Parameters > Excel upload |
| Reset parameter datasets when the uploaded file is absent | 131 | 137 | User Supplied Parameters > Mass List / Reference Mass List / Parameters > Excel upload |
| Add a metabolite row and reset its input fields | 141 | 191 | User Supplied Parameters > Mass List |
| Delete selected metabolite row | 194 | 199 | User Supplied Parameters > Mass List |
| Clear metabolite list | 201 | 203 | User Supplied Parameters > Mass List |
| Display metabolite table | 205 | 206 | User Supplied Parameters > Mass List |
| Add a reference-mass row and reset its input fields | 209 | 257 | User Supplied Parameters > Reference Mass List |
| Delete selected reference-mass row | 260 | 266 | User Supplied Parameters > Reference Mass List |
| Clear reference-mass list | 268 | 270 | User Supplied Parameters > Reference Mass List |
| Display reference-mass table | 272 | 273 | User Supplied Parameters > Reference Mass List |
| Add a processing-parameter row and reset its input fields | 276 | 304 | User Supplied Parameters > Parameters |
| Delete selected processing-parameter row | 307 | 313 | User Supplied Parameters > Parameters |

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Clear processing parameters | 315 | 317 | User Supplied Parameters > Parameters |
| Display processing-parameter table | 319 | 320 | User Supplied Parameters > Parameters |
| Initialize project-information state | 326 | 329 | User Supplied Parameters > Project Information |
| Store project information entered by the user | 331 | 336 | User Supplied Parameters > Project Information |
| Display stored project information | 338 | 343 | User Supplied Parameters > Project Information |
| Clear project information and its input fields | 346 | 358 | User Supplied Parameters > Project Information |
| Read uploaded migration-index Excel file | 363 | 369 | User Supplied Parameters > User Supplied Migration Indexes |
| Add a migration-index row and reset its input fields | 372 | 405 | User Supplied Parameters > User Supplied Migration Indexes |
| Delete selected migration-index row | 408 | 416 | User Supplied Parameters > User Supplied Migration Indexes |
| Clear migration-index list | 419 | 422 | User Supplied Parameters > User Supplied Migration Indexes |
| Display migration-index table | 425 | 426 | User Supplied Parameters > User Supplied Migration Indexes |
| Download current mass lists and parameters as Excel | 431 | 445 | User Supplied Parameters > Save Table (.xlsx) (shared download button) |

## 3. Engine: file selection and upload

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Initialize uploaded-file and pseudo-file state | 451 | 454 | Engine > real/pseudo-file lists (shared state; Clear Files) |
| Register uploaded mz5 files, display their names, and optionally trigger autorun | 456 | 479 | Engine > File Uploads for Initializing Data Preprocessing > Upload .mz5 Files / autorun |
| Clear uploaded-file and pseudo-file lists | 482 | 486 | Engine > real/pseudo-file lists (shared state; Clear Files) |
| Display editable pseudo-file-name table | 491 | 493 | Engine > Generate Pseudo .mz5 Files to Access Stored Datafiles |
| Apply edits to pseudo-file names | 496 | 500 | Engine > Generate Pseudo .mz5 Files to Access Stored Datafiles |
| Read pseudo-file names from Excel | 503 | 510 | Engine > Generate Pseudo .mz5 Files to Access Stored Datafiles |
| Generate empty pseudo-mz5 files and register them | 514 | 533 | Engine > Generate Pseudo .mz5 Files to Access Stored Datafiles |
| Add an empty pseudo-file-name row | 536 | 541 | Engine > Generate Pseudo .mz5 Files to Access Stored Datafiles |

## 4. Initialize Run: preparation and timing

The complete Initialize Run handler spans lines 546–2531. Its per-file loop spans lines 787–2521.

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Require an uploaded mz5 file | 547 | 549 | Engine > Initialize Run > run setup / processing (background) |
| Create a uniquely named Results folder | 551 | 557 | Engine > Initialize Run > run setup / processing (background) |
| Define timing recorder, stage transitions, CSV persistence, and total timing | 559 | 604 | Engine > Initialize Run > progress messages + run_timings.csv (background) |
| Initialize run timing and connect stage messages to progress display | 605 | 610 | Engine > Initialize Run > progress messages + run_timings.csv (background) |
| Create Plots and Data directories | 612 | 616 | Engine > Initialize Run > run setup / processing (background) |

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Collect target names, reject duplicates, and determine target/injection counts | 618 | 631 | Engine > Initialize Run > run setup / processing (background) |
| Create target-specific plot directories when using Metabolite format | 633 | 638 | Engine > Initialize Run; uses User Supplied Parameters > Parameters > plot.format |
| Save a copy of the input mass lists and parameters | 640 | 647 | Engine > Initialize Run > run setup / processing (background) |
| Assemble supplied metabolite and reference migration-time tables | 651 | 654 | Engine > Initialize Run; uses User Supplied Parameters > Mass List + Reference Mass List |
| Identify the reference standard preceding each metabolite | 656 | 671 | Engine > Initialize Run; uses User Supplied Parameters > Mass List + Reference Mass List |
| Identify the reference standard following each metabolite | 673 | 688 | Engine > Initialize Run; uses User Supplied Parameters > Mass List + Reference Mass List |
| Calculate migration indexes or relative migration times | 690 | 732 | Engine > Initialize Run; uses User Supplied Parameters > Mass List + Reference Mass List |
| Assemble the migration-index summary table | 734 | 740 | Engine > Initialize Run; uses User Supplied Parameters > Mass List + Reference Mass List |
| Validate and apply user-supplied migration-index overrides | 743 | 766 | Engine > Initialize Run; uses User Supplied Parameters > User Supplied Migration Indexes / Parameters > Manual.Indexes |
| Export migration-index summary CSV | 768 | 771 | Engine > Initialize Run > run setup / processing (background) |
| Collect uploaded file paths/names and initialize Plotly collection | 774 | 777 | Engine > Initialize Run > run setup / processing (background) |
| Define helper for saving collected Plotly objects | 781 | 783 | Engine > Initialize Run > run setup / processing (background) |
| Start per-file progress reporting and create the file’s Data directory | 787 | 796 | Engine > Initialize Run > run setup / processing (background) |

## 5. Initialize Run: raw data to eie_df

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Create temporary mz5 copy | 799 | 802 | Engine > Initialize Run > mz5 reading/spectrum preparation (background) |
| Read mz5 spectra into run_data and unlock its assay-data environment | 804 | 823 | Engine > Initialize Run > mz5 reading/spectrum preparation (background) |
| Read calibration settings and validate the Yes/No choice | 825 | 848 | Engine > Initialize Run > mass calibration (background); settings: User Supplied Parameters > Parameters |
| Estimate a reference mass's measured position and mass error — existing calibration_parameters() | 868 | 909 | Engine > Initialize Run > mass calibration (background); settings: User Supplied Parameters > Parameters |
| Iterate through spectra and obtain both reference-mass measurements | 913 | 941 | Engine > Initialize Run > mass calibration (background); settings: User Supplied Parameters > Parameters |
| Fit mass-error correction model | 943 | 948 | Engine > Initialize Run > mass calibration (background); settings: User Supplied Parameters > Parameters |
| Apply correction to spectrum m/z values | 950 | 958 | Engine > Initialize Run > mass calibration (background); settings: User Supplied Parameters > Parameters |
| Calculate target extraction windows from m/z and ppm tolerance | 978 | 986 | Engine > Initialize Run > target extraction (background); settings: both mass lists |
| Inspect spectra, handle invalid m/z values, and rebuild sorted spectra | 988 | 1045 | Engine > Initialize Run > mz5 reading/spectrum preparation (background) |
| Extract electropherograms using chromatogram() | 1047 | 1054 | Engine > Initialize Run > target extraction (background); settings: both mass lists |
| Assemble time and target-intensity columns into eie_df | 1056 | 1065 | Engine > Initialize Run > target extraction (background); settings: both mass lists |
| Validate smoothing choice and optionally smooth each intensity column | 1067 | 1113 | Engine > Initialize Run > smoothing (background); settings: mass lists + Parameters |

*The complete optional calibration block is lines 825–968. The extraction-to-processing handoff is after line 1065, before optional smoothing.*

## 6. Initialize Run: internal-standard peak processing

The complete internal-standard processing stage spans lines 1115–1443.

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Detect candidate peaks and collect boundaries, apexes, and intensities | 1125 | 1165 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Filter candidates by allowed migration-time range | 1167 | 1171 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Integrate candidate peaks and subtract baseline area | 1173 | 1201 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Name peak columns and retain candidates for later recovery | 1203 | 1215 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Calculate full width at half maximum and filter by width tolerance | 1217 | 1256 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Retain peaks with the largest areas up to the injection-count cutoff | 1258 | 1262 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Calculate spacing limits and identify abnormal peak gaps | 1264 | 1289 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Correct one abnormal gap, recovering a candidate or creating a placeholder | 1291 | 1367 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Correct two or three abnormal gaps | 1369 | 1420 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Combine processed internal-standard peak tables | 1423 | 1434 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |
| Extract standard apex times for metabolite-time prediction | 1436 | 1443 | Engine > Initialize Run > internal-standard processing (background); settings: User Supplied Parameters > Reference Mass List |

## 7. Initialize Run: metabolite peak processing

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Detect candidate metabolite peaks and collect boundaries/intensities | 1455 | 1501 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Filter candidates by minimum peak width | 1503 | 1509 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Integrate candidates, subtract baseline, and name peak columns | 1511 | 1547 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Predict migration times from indexes and detected reference standards | 1549 | 1576 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Assign peaks within migration windows; create placeholders when absent | 1578 | 1621 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Create placeholders for predictions beyond the recorded run time | 1623 | 1647 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Resolve duplicate peak assignments with progressively narrower windows | 1650 | 1723 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Correct assignments using expected spacing between injections | 1725 | 1863 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Combine processed metabolite peak tables | 1865 | 1877 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Estimate noise thresholds and annotate <LOD and NPD peaks | 1879 | 1932 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Detect configured interferences and annotate overlapping peaks | 1934 | 1986 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |
| Combine standard/metabolite peaks and their comment tables | 1988 | 2002 | Engine > Initialize Run > metabolite processing (background); settings: User Supplied Parameters > Mass List / Migration Indexes |

*Within the noise stage, lines 1906–1908 already define the small noise_calculation() function.*

## 8. Initialize Run: plotting and result export

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Initialize per-target plot-data collection | 2010 | 2013 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Build peak annotations | 2019 | 2024 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Build integration shading data and assign peak numbers | 2026 | 2053 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Add baseline values to integration shading | 2055 | 2070 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Assemble per-target traces, annotations, labels, and axis limits | 2072 | 2084 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Define static electropherogram plot — plot_function() | 2088 | 2123 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Define interactive electropherogram plot — plot_function_plotly() | 2126 | 2182 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Create plot directories for Sample format | 2184 | 2207 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Save internal-standard PNGs in Sample format | 2212 | 2233 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Save metabolite/reference composite PNGs for relative-time targets | 2235 | 2276 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Save composite PNGs for migration-index targets | 2278 | 2308 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Save PNGs organized by metabolite | 2310 | 2336 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Refresh final target’s stored plot-data entry | 2338 | 2347 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Build and collect internal-standard Plotly objects | 2350 | 2368 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Build and collect metabolite Plotly objects | 2370 | 2418 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Publish collected plots to reactive state | 2420 | 2426 | Engine > Initialize Run > plot generation/export (background); later displayed in Visualization |
| Build peak-area report, apply annotations, and accumulate files | 2429 | 2449 | Engine > Initialize Run > result/file export (background); later used by Visualization / Downstream Processing |
| Build migration-time report and accumulate files | 2452 | 2463 | Engine > Initialize Run > result/file export (background); later used by Visualization / Downstream Processing |
| Save per-file plot_list_data.RData | 2465 | 2467 | Engine > Initialize Run > result/file export (background); later used by Visualization / Downstream Processing |
| Open/initialize SQLite, replace existing file records, and write results | 2469 | 2496 | Engine > Initialize Run > SQLite export (background); later used by Visualization |
| Remove temporary mz5 copy | 2499 | 2504 | Engine > Initialize Run > result/file export (background); later used by Visualization / Downstream Processing |
| Export accumulated peak-area and migration-time CSVs | 2506 | 2514 | Engine > Initialize Run > result/file export (background); later used by Visualization / Downstream Processing |
| Save accumulated Plotly objects and finish file-stage timing | 2516 | 2519 | Engine > Initialize Run > result/file export (background); later used by Visualization / Downstream Processing |
| Finalize run timing or record failure/interruption | 2523 | 2530 | Engine > Initialize Run > run_timings.csv / failure reporting (background) |

## 9. Visualization: state and database helpers

These tasks already contain several named functions.

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Initialize current results, working metadata, and undo state | 2538 | 2568 | Visualization > loaded results / editing / undo state (background) |
| Resolve selected Results directory — main_folder | 2571 | 2574 | Visualization > shared controls > Select Results Folder |
| Load selected file's .RData results — plot_list_data | 2577 | 2590 | Visualization > legacy .RData reactive helper (no consumer found) |
| Query available target/plot names — filtered_plot_names | 2594 | 2608 | Visualization > Loaded Plots / Dual EIE View selection tables |
| Track modified plots and render invalidation | 2611 | 2615 | Visualization > Edited plots to be regenerated / plot refresh |
| Construct SQLite path — get_db_path() | 2621 | 2623 | Shared database support: Engine > Initialize Run / Visualization > folder loading, migration, editing (background) |
| Check database existence — db_exists() | 2626 | 2628 | Shared database support: Engine > Initialize Run / Visualization > folder loading, migration, editing (background) |
| Open database with write-ahead logging — open_db() | 2631 | 2636 | Shared database support: Engine > Initialize Run / Visualization > folder loading, migration, editing (background) |
| Create database tables and indexes — initialize_db() | 2639 | 2722 | Shared database support: Engine > Initialize Run / Visualization > folder loading, migration, editing (background) |
| Safely obtain a column, supplying missing values — safe_col() | 2725 | 2727 | Shared database support: Engine > Initialize Run / Visualization > folder loading, migration, editing (background) |
| Write one file's results transactionally — write_file_to_db() | 2730 | 2871 | Shared database support: Engine > Initialize Run / Visualization > folder loading, migration, editing (background) |
| Read one target's trace from .RData and editable data from SQLite — read_plot_data_from_db() | 2874 | 2941 | Visualization > file/plot loading, Plotting, Dual EIE View (background) |
| Record an edit audit entry — log_edit() | 2944 | 2950 | Visualization > Plotting > peak-edit actions > edit log (background) |

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Migrate existing .RData results into SQLite — migrate_rdata_to_db() | 2953 | 3013 | Visualization > Select Results Folder > Database Not Found > Yes — build database |
| Reconstruct wide result tables from SQLite — load_wide_data_from_db() | 3017 | 3108 | Visualization > file/plot loading, Plotting, Dual EIE View (background) |
| Assemble a file's complete plot-data collection — load_plot_list_from_db() | 3111 | 3121 | Visualization > file/plot loading, Plotting, Dual EIE View (background) |

## 9a. Database writer: individual storage tasks

These are subtasks within write_file_to_db(), listed in section 9.

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Transform and store annotations | 2736 | 2754 | Shared storage: Engine > Initialize Run; Visualization > Select Results Folder > Build database (background) |
| Transform and store integration shading/baselines | 2756 | 2774 | Shared storage: Engine > Initialize Run; Visualization > Select Results Folder > Build database (background) |
| Store axes and plot labels | 2776 | 2790 | Shared storage: Engine > Initialize Run; Visualization > Select Results Folder > Build database (background) |
| Convert peak measurements from wide to long format and store them | 2792 | 2809 | Shared storage: Engine > Initialize Run; Visualization > Select Results Folder > Build database (background) |
| Convert peak areas to long format and store them | 2811 | 2828 | Shared storage: Engine > Initialize Run; Visualization > Select Results Folder > Build database (background) |
| Convert migration times to long format and store them | 2830 | 2843 | Shared storage: Engine > Initialize Run; Visualization > Select Results Folder > Build database (background) |
| Serialize and store migration indexes | 2845 | 2864 | Shared storage: Engine > Initialize Run; Visualization > Select Results Folder > Build database (background) |
| Commit or roll back the write transaction | 2866 | 2870 | Shared storage: Engine > Initialize Run; Visualization > Select Results Folder > Build database (background) |

## 10. Visualization: selecting results and displaying plots

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Populate source-file selector from uploaded files | 3127 | 3129 | Visualization > shared controls > Select File / Select Results Folder |
| Discover Results directories — list_results_folders() | 3133 | 3137 | Visualization + Downstream Processing > Select Results Folder choices (background) |
| Poll for new Results directories | 3140 | 3147 | Visualization + Downstream Processing > Select Results Folder choices (background) |
| Update Results-directory selector | 3150 | 3155 | Visualization > shared controls > Select Results Folder |
| Initialize database connection state | 3158 | 3158 | Visualization > shared controls > Select Results Folder |
| Switch Results directory: close old database, open selected database, or offer migration | 3161 | 3203 | Visualization > shared controls > Select Results Folder |
| Handle migration confirmation, progress, outcome, and connection | 3206 | 3256 | Visualization > Database Not Found > Yes — build database dialog |
| Close database when the Shiny session ends | 3259 | 3264 | Visualization > session close > database cleanup (background) |
| Load selected file’s database results into reactive state | 3267 | 3321 | Visualization > shared controls > Select File / Select Results Folder |
| Load results when a database connection becomes available | 3324 | 3379 | Visualization > shared controls > Select File / Select Results Folder |
| Display available plot names | 3386 | 3388 | Visualization > shared controls > Loaded Plots |
| Load and render selected electropherogram, preserving zoom | 3392 | 3430 | Visualization > Plotting > selected electropherogram |
| Initialize and switch between zoom and rectangle-drawing modes | 3436 | 3455 | Visualization > Plotting > Draw Mode |
| Initialize zoom state | 3458 | 3461 | Visualization > Plotting > plot zoom / selection |
| Capture zoom changes and reset on autorange | 3464 | 3479 | Visualization > Plotting > plot zoom / selection |
| Reset zoom when selecting a different target | 3482 | 3487 | Visualization > Plotting > plot zoom / selection |
| Capture dragged integration lines or rectangle-selected boundaries | 3490 | 3548 | Visualization > Plotting > red/blue lines and Draw Mode boundaries |

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Display integration boundary positions | 3552 | 3557 | Visualization > Plotting > red/blue lines and Draw Mode boundaries |
| Query and display selected target’s peak annotations | 3561 | 3583 | Visualization > Plotting > Peak Position Information |

## 11. Visualization: editing and saving peaks

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Load .RData results for editing — loaded_plot_list_data | 3590 | 3608 | Visualization > Select File > .RData loading/edit-state initialization |
| Persist a target's edits to SQLite and .RData, then refresh state — save_plot_data() | 3611 | 3716 | Visualization > Plotting > all peak-edit actions (background persistence/reload) |
| Reload result tables from SQLite — reload_plot_data() | 3720 | 3736 | Visualization > Plotting > all peak-edit actions (background persistence/reload) |
| Initialize editing state from .RData when switching files | 3741 | 3760 | Visualization > Select File > .RData loading/edit-state initialization |
| Mark selected peaks as not detected and remove their integration shading | 3763 | 3847 | Visualization > Plotting > Delete Selected Peaks |
| Mark all interior peaks as not detected, retaining first and last | 3851 | 3930 | Visualization > Plotting > Delete Middle Peaks (Keeps first and last) |
| Mark every peak as not detected | 3933 | 4008 | Visualization > Plotting > Delete all peaks |
| Restore previous metadata and save it to .RData | 4012 | 4056 | Visualization > Plotting > Undo Peak Deletion |
| Show peak-number input dialog — peak_input_modal() | 4566 | 4575 | Visualization > Plotting > Manually Adjust Integration / Manual Peak Input dialog |
| Integrate between selected boundaries and update measurements — manual_peak_integration() | 4579 | 4708 | Visualization > Plotting > Manually Adjust Integration / Manual Peak Input dialog |
| Handle manual integration using table selection or open the dialog | 4712 | 4734 | Visualization > Plotting > Manually Adjust Integration / Manual Peak Input dialog |
| Handle manually entered peak number and integrate | 4737 | 4757 | Visualization > Plotting > Manually Adjust Integration / Manual Peak Input dialog |
| Initialize baseline-adjustment modes | 4761 | 4762 | Visualization > Plotting > Adjust Individual Baseline / Adjust All Baselines modes |
| Show peak-number dialog for baseline adjustment | 4765 | 4772 | Visualization > Plotting > Adjust Individual Baseline / peak-number dialog / plot click |

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Recalculate one peak using a clicked baseline — adjust_individual_baseline() | 4775 | 4882 | Visualization > Plotting > Adjust Individual Baseline / peak-number dialog / plot click |
| Activate individual-baseline adjustment | 4886 | 4891 | Visualization > Plotting > Adjust Individual Baseline / peak-number dialog / plot click |
| Capture baseline click and apply/request peak selection | 4894 | 4918 | Visualization > Plotting > Adjust Individual Baseline / peak-number dialog / plot click |
| Handle manually entered peak number for baseline adjustment | 4921 | 4924 | Visualization > Plotting > Adjust Individual Baseline / peak-number dialog / plot click |
| Show whole-plot baseline dialog — show_adjust_all_baselines_modal() | 4929 | 4939 | Visualization > Plotting > Adjust All Baselines dialog / plot click |
| Recalculate all peaks using a common baseline — adjust_plot_baseline() | 4942 | 5018 | Visualization > Plotting > Adjust All Baselines dialog / plot click |
| Activate whole-plot baseline adjustment | 5022 | 5028 | Visualization > Plotting > Adjust All Baselines dialog / plot click |
| Render electropherogram inside baseline dialog | 5031 | 5058 | Visualization > Plotting > Adjust All Baselines dialog / plot click |
| Capture clicked baseline and apply it to all peaks | 5061 | 5078 | Visualization > Plotting > Adjust All Baselines dialog / plot click |
| Close whole-plot baseline dialog | 5081 | 5083 | Visualization > Plotting > Adjust All Baselines dialog / plot click |
| Display three target-selection tables for EIE comparison | 5087 | 5097 | Visualization > Dual EIE View > Select Plots to Compare / Combined Electropherograms |
| Load and display two or three stacked EIE plots | 5101 | 5150 | Visualization > Dual EIE View > Select Plots to Compare / Combined Electropherograms |
| Mark selected peaks as interfered and save the changes | 5153 | 5232 | Visualization > Plotting > Mark Peak Interference |

## 11a. Edit saving: individual persistence tasks

These are subtasks within save_plot_data(), listed in section 11.

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Replace target annotations in SQLite | 3617 | 3632 | Visualization > Plotting > peak-edit actions (background save/refresh) |
| Replace target integration data in SQLite | 3634 | 3652 | Visualization > Plotting > peak-edit actions (background save/refresh) |
| Replace target peak measurements in SQLite | 3654 | 3674 | Visualization > Plotting > peak-edit actions (background save/refresh) |
| Replace target peak areas in SQLite | 3676 | 3687 | Visualization > Plotting > peak-edit actions (background save/refresh) |
| Append edit-log entry | 3689 | 3690 | Visualization > Plotting > peak-edit actions (background save/refresh) |
| Save .RData backup | 3692 | 3703 | Visualization > Plotting > peak-edit actions (background save/refresh) |
| Reload target data and trigger display refresh | 3705 | 3715 | Visualization > Plotting > peak-edit actions (background save/refresh) |

## 12. Visualization: rebuilding plots and reports

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Define static electropherogram plot for regeneration — plot_function() | 4062 | 4099 | Visualization > Plotting > Other > Regenerate Changed Plots > PNG builder (background) |
| Define interactive electropherogram plot for visualization — plot_function_plotly() | 4102 | 4158 | Visualization > Plotting / Dual EIE View / baseline dialog > shared plot builder |
| Rebuild peak-area CSV from SQLite — regenerate_metabolite_peak_areas() | 4161 | 4187 | Visualization > Plotting > Other > Rebuild Peak Area CSV; also called by integration/regeneration |
| Rebuild and compress all saved Plotly objects — regenerate_plots() | 4190 | 4244 | Visualization > saved Plotly rebuild helper (defined; no button caller found) |
| Regenerate changed PNGs in Sample or Metabolite layout — regenerate_ggplot() | 4247 | 4403 | Visualization > Plotting > Other > Regenerate Changed Plots |
| Rebuild migration-time CSV from annotations — regenerate_metabolite_migration_times() | 4406 | 4432 | Visualization > Plotting > Other > Rebuild Peak Migration Time CSV |
| Define loading-screen message collection | 4437 | 4468 | Visualization > Plotting > Other > regeneration loading screen; shared with Drug Analysis > parsing |
| Display additional-options dialog | 4471 | 4486 | Visualization > Plotting > Other dialog |
| Coordinate changed-plot regeneration, loading screen, and peak-area export | 4489 | 4535 | Visualization > Plotting > Other > Regenerate Changed Plots |
| Display list of modified plots | 4538 | 4540 | Visualization > shared controls > Edited plots to be regenerated |
| Handle standalone peak-area CSV rebuild | 4543 | 4551 | Visualization > Plotting > Other > Rebuild Peak Area CSV |
| Handle standalone migration-time CSV rebuild | 4554 | 4562 | Visualization > Plotting > Other > Rebuild Peak Migration Time CSV |

*regenerate_plots() is defined, but the current regeneration button calls the peak-area and PNG regeneration functions; it does not call this Plotly rebuild helper.*

## 13. Downstream processing: data loading and transformation

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Read peak-area and migration-time CSVs — grab_metabolite_files() | 5237 | 5252 | Downstream Processing > shared controls > Set working environment and initialize data |
| Initialize downstream datasets | 5255 | 5256 | Downstream Processing > shared controls > Set working environment and initialize data |
| Load selected Results folder's CSVs into state | 5259 | 5266 | Downstream Processing > shared controls > Set working environment and initialize data |
| Resolve downstream Results folder | 5269 | 5272 | Downstream Processing > shared controls > Set working environment and initialize data |
| Populate downstream Results/file selectors | 5275 | 5278 | Downstream Processing > shared controls > Select Results Folder / Select File |
| Parse m/z and name from target identifiers — extract_mass() | 5283 | 5287 | Downstream Processing > m/z vs Migration Time + Downstream Processing > Data Quality & Variability > CV plot > parse target names (background) |
| Assemble m/z-versus-migration-time data for selected file/injection | 5290 | 5323 | Downstream Processing > m/z vs Migration Time; controls: shared Select File / Select Peak Number |
| Display interactive m/z-versus-migration-time plot | 5326 | 5338 | Downstream Processing > m/z vs Migration Time; controls: shared Select File / Select Peak Number |
| Display injection/peak-number selector | 5341 | 5344 | Downstream Processing > m/z vs Migration Time; controls: shared Select File / Select Peak Number |
| Show normalization-reference selection dialog | 5349 | 5359 | Downstream Processing > shared controls > Editing your data matrix > Normalize Data > Normalization Options dialog |
| Validate, normalize, update state, and save normalized areas | 5362 | 5405 | Downstream Processing > shared controls > Editing your data matrix > Normalize Data > Normalization Options dialog |
| Perform row-wise normalization — nested normalize_peak_areas() | 5377 | 5394 | Downstream Processing > shared controls > Editing your data matrix > Normalize Data > Normalization Options dialog |

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Replace missing values with minimum/5 or zero, then save | 5409 | 5443 | Downstream Processing > shared controls > Editing your data matrix > missing-data method / Replace Missing Data |
| Read metadata, match file/injection to sample IDs, merge, and save | 5449 | 5509 | Downstream Processing > shared controls > Connecting metadata > Connect Metadata to Results |
| Extract one file’s block of peak-area rows — nested filter_data_by_file_name() | 5487 | 5491 | Downstream Processing > shared controls > Connecting metadata > Connect Metadata to Results |

## 14. Downstream processing: quality control and statistics

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Populate control-chart reference selector | 5514 | 5519 | Downstream Processing > shared controls > Generate Control Charts > Select Metabolite |
| Generate raw-area control chart with mean and ±3 SD limits | 5522 | 5550 | Downstream Processing > Control Charts > Control chart for non-normalized data |
| Generate processed-area control chart with mean and ±3 SD limits | 5554 | 5575 | Downstream Processing > Control Charts > Control chart for normalized data (current in-memory matrix) |
| Calculate per-column coefficient of variation — calculate_cv() | 5582 | 5585 | Downstream Processing > Data Quality & Variability > Calculate CVs > CV calculation (background) |
| Define reactive CV calculation for current area data — cv_data | 5588 | 5593 | Downstream Processing > Data Quality & Variability > legacy reactive statistics helper (no consumer found) |
| Perform Pareto scaling — pareto_scale() | 5597 | 5600 | Downstream Processing > Data Quality & Variability > PCA preprocessing/calculation (background) |
| Remove zero-variance columns — remove_zero_variance() | 5603 | 5608 | Downstream Processing > Data Quality & Variability > PCA preprocessing/calculation (background) |
| Validate data, scale, and calculate PCA — perform_pca() | 5611 | 5636 | Downstream Processing > Data Quality & Variability > PCA preprocessing/calculation (background) |
| Define reactive PCA calculation for current data — pca_data | 5639 | 5645 | Downstream Processing > Data Quality & Variability > legacy reactive statistics helper (no consumer found) |
| Parse QC categories/exclusions, split samples, filter invalid columns, calculate CVs and PCA | 5649 | 5714 | Downstream Processing > Data Quality & Variability > QC identifier / exclusions / Calculate CVs |
| Calculate and display technical/biological CV summary table | 5721 | 5738 | Downstream Processing > Data Quality & Variability > Average CVs (%) |
| Prepare and display CV-versus-m/z plot | 5740 | 5765 | Downstream Processing > Data Quality & Variability > CV based on m/z of metabolite |

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Build, save, and display QC/non-QC PCA plot | 5767 | 5808 | Downstream Processing > Data Quality & Variability > PCA plot + saved QC_PCA_plot.png |
| Display current downstream peak-area table | 5814 | 5824 | Downstream Processing > Metadata > Peak Area Datatable |

*The compute_cv button handler enclosing the CV table and both plots spans lines 5718–5809.*

## 15. Downstream processing: batch correction

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Install/load optional batch-correction packages | 5828 | 5849 | Downstream Processing > Batch Correction > Initialize Packages |
| Read uploaded batch-order CSV | 5852 | 5855 | Downstream Processing > Batch Correction > Upload Batch Correction Order |
| Obtain area data and batch metadata; log-transform metabolite columns | 5861 | 5872 | Downstream Processing > Batch Correction > Perform Batch Correction (file output; no result panel) |
| Configure correction strategies and reference-sample groups | 5874 | 5889 | Downstream Processing > Batch Correction > Perform Batch Correction (file output; no result panel) |
| Apply batch correction to each metabolite | 5891 | 5901 | Downstream Processing > Batch Correction > Perform Batch Correction (file output; no result panel) |
| Name correction results and reverse log transformation | 5904 | 5910 | Downstream Processing > Batch Correction > Perform Batch Correction (file output; no result panel) |
| Export transformed and original-scale corrected results | 5912 | 5914 | Downstream Processing > Batch Correction > Perform Batch Correction (file output; no result panel) |

*The complete batch-correction button handler spans lines 5859–5916.*

## 16. Reporting: MetaboAnalyst export

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Initialize reporting-matrix state | 5920 | 5920 | Reporting > Generate Data Matrix for MetaboAnalyst |
| Validate input data and identify output folder | 5925 | 5930 | Reporting > Generate Data Matrix for MetaboAnalyst |
| Remove file/peak identifiers, transpose data, and obtain sample IDs | 5932 | 5936 | Reporting > Generate Data Matrix for MetaboAnalyst |
| Parse user-defined sample categories | 5938 | 5943 | Reporting > Specify classes for MetaboAnalyst matrix |
| Assign a class to each sample | 5945 | 5963 | Reporting > Specify classes for MetaboAnalyst matrix |
| Insert and order sample-ID and CLASS rows | 5965 | 5970 | Reporting > Generate Data Matrix for MetaboAnalyst |
| Store and export MetaboAnalyst matrix | 5972 | 5979 | Reporting > Generate Data Matrix for MetaboAnalyst |
| Display generated matrix | 5984 | 5990 | Reporting > MetaboAnalyst Matrix |

*The complete matrix-generation handler spans lines 5923–5980.*

## 17. Drug Analysis: data import

| Task / existing function | Start | End | GUI location / trigger |
|---|---|---|---|
| Initialize patient, DrugBank, and TWOSIDES state | 5999 | 6001 | Drug Analysis > patient and drug database state (background) |
| Read uploaded adherence-results Excel file | 6005 | 6010 | Drug Analysis > Upload Adherence Results (Excel) |
| Load TWOSIDES database | 6015 | 6017 | Drug Analysis > Load Drug Databases / Database Not Found dialog |
| Load cached DrugBank data or offer XML parsing | 6019 | 6040 | Drug Analysis > Load Drug Databases / Database Not Found dialog |
| Show loading screen while parsing DrugBank | 6042 | 6069 | Drug Analysis > DrugBank Database Not Found > Yes, parse now dialog |
| Select DrugBank fields and parse XML database | 6071 | 6083 | Drug Analysis > DrugBank Database Not Found > Yes, parse now dialog |
| Cache parsed DrugBank data, update state, and report completion | 6085 | 6092 | Drug Analysis > DrugBank Database Not Found > Yes, parse now dialog |

*This final section currently imports data; it does not yet calculate drug adherence.*

## Wiring notes

A visible control is not proof of an implemented action. The GUI guide identifies the unhandled .RData upload/reset widgets, the unbound progressBar output slot, duplicate input IDs, reference/manual-index field mismatches, incomplete database-backed undo, and migration-time export unit inconsistency. These findings are documented; application behavior was not changed.

## Structural observation

Calculations, file/database access, reactive-state updates and UI actions are mixed within blocks. Existing helpers often read Shiny state directly. Choosing reusable functions will require explicit inputs/outputs and preserving the GUI triggers mapped here.